

## FURTHER MECHANICS / CENTRE OF MASS

# Centre of Mass of Systems of Particles

Worked solutions

9 questions · 57 marks

**Instructions**

Unless a question states otherwise, take  $g = 9.8 \text{ m s}^{-2}$  wherever a numerical value of  $g$  is required. Solutions also give final answers for  $g = 10 \text{ m s}^{-2}$  where these differ.

## Questions

|            |          |    |
|------------|----------|----|
| Question 1 | 6 marks  | 2  |
| Question 2 | 11 marks | 4  |
| Question 3 | 7 marks  | 7  |
| Question 4 | 5 marks  | 9  |
| Question 5 | 4 marks  | 11 |
| Question 6 | 7 marks  | 12 |
| Question 7 | 6 marks  | 13 |
| Question 8 | 5 marks  | 15 |
| Question 9 | 6 marks  | 16 |

1. Three particles of masses  $m$ ,  $2m$  and  $km$  are placed at the points  $A(a, 0)$ ,  $B(-2a, 3a)$  and  $C(2a, -a)$  respectively, where  $k > 0$ .

The centre of mass of the three particles is the point with coordinates  $(\bar{x}, \bar{y})$ .

- (a) (i) Find  $\bar{x}$  in terms of  $a$  and  $k$ .  
 (ii) Find  $\bar{y}$  in terms of  $a$  and  $k$ .

[4]

- (b) Given that the centre of mass lies on the circle with centre  $(a, 0)$  and radius  $\frac{\sqrt{2}}{2}a$ , find the possible values of  $k$ .

[2]

### Solution

- (a) The total mass is

$$M = m + 2m + km = (k + 3)m$$

- (i) Using the formula for the  $x$ -coordinate of the centre of mass,

$$M\bar{x} = m(a) + 2m(-2a) + km(2a)$$

So

$$\begin{aligned}(k + 3)m\bar{x} &= am - 4am + 2kam \\ &= am(2k - 3)\end{aligned}$$

Hence

$$\bar{x} = \frac{a(2k - 3)}{k + 3}$$

- (ii) Using the formula for the  $y$ -coordinate of the centre of mass,

$$M\bar{y} = m(0) + 2m(3a) + km(-a)$$

So

$$\begin{aligned}(k + 3)m\bar{y} &= 0 + 6am - kam \\ &= am(6 - k)\end{aligned}$$

Hence

$$\bar{y} = \frac{a(6 - k)}{k + 3}$$

- (b) Since the centre of mass lies on the circle with centre  $(a, 0)$  and radius  $\frac{\sqrt{2}}{2}a$ ,

$$(\bar{x} - a)^2 + \bar{y}^2 = \left(\frac{\sqrt{2}}{2}a\right)^2 = \frac{a^2}{2}$$

From part (a),

$$\begin{aligned}\bar{x} - a &= \frac{a(2k - 3)}{k + 3} - a \\ &= a\left(\frac{2k - 3 - (k + 3)}{k + 3}\right) \\ &= \frac{a(k - 6)}{k + 3}\end{aligned}$$

and

$$\bar{y} = \frac{a(6 - k)}{k + 3} = -\frac{a(k - 6)}{k + 3}$$

Substituting into the circle equation,

$$\left(\frac{a(k-6)}{k+3}\right)^2 + \left(-\frac{a(k-6)}{k+3}\right)^2 = \frac{a^2}{2}$$
$$2\left(\frac{a(k-6)}{k+3}\right)^2 = \frac{a^2}{2}$$

So

$$4(k-6)^2 = (k+3)^2$$

Expanding,

$$4(k^2 - 12k + 36) = k^2 + 6k + 9$$

$$4k^2 - 48k + 144 = k^2 + 6k + 9$$

$$3k^2 - 54k + 135 = 0$$

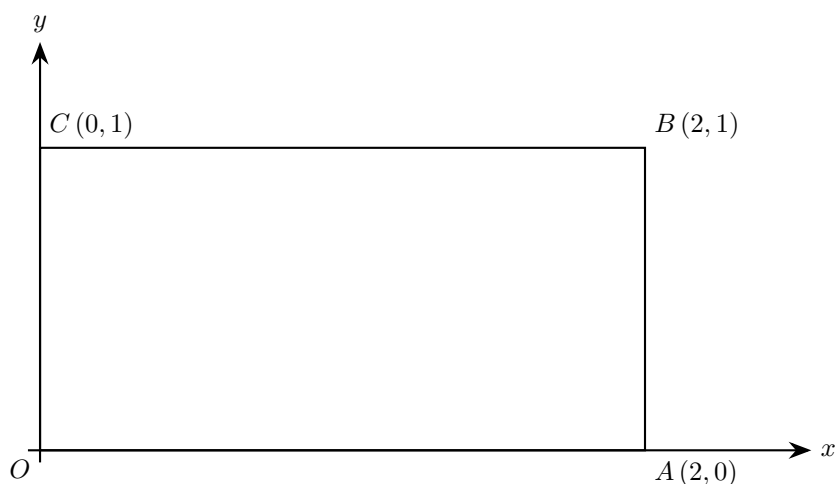
$$k^2 - 18k + 45 = 0$$

Factorising,

$$(k-3)(k-15) = 0$$

Hence the possible values of  $k$  are

$$k = 3 \quad \text{or} \quad k = 15$$



2. The diagram shows a light rectangular lamina  $OABC$  where  $O$  is the origin of the coordinate system in which the points  $A$ ,  $B$  and  $C$  have coordinates  $(2, 0)$ ,  $(2, 1)$  and  $(0, 1)$  respectively. Coordinates refer to the axes shown and the units are metres.

The following three forces are applied to the lamina.

$$\begin{pmatrix} p \\ 2q \end{pmatrix} \text{ N at the point } (1, 0)$$

$$\begin{pmatrix} -q \\ p \end{pmatrix} \text{ N at the point } C$$

$$\begin{pmatrix} -3 \\ -6 \end{pmatrix} \text{ N at the point } (t, 1), \text{ where } 0 \leq t \leq 2$$

The three forces form a couple of magnitude  $3 \text{ N m}$  in the anticlockwise direction.

- (a) Determine the values of  $p$ ,  $q$  and  $t$ .

[5]

The three forces are removed. The lamina is in the same position as shown in the diagram.

Three particles are now attached to certain points on the lamina as follows.

- (b) Find the coordinates of the composite body's centre of mass. Give your answer in terms of  $\lambda$ , where the particles are
- a particle of mass  $2 \text{ kg}$  at the point  $(2\lambda^2, \lambda)$ , where  $0 \leq \lambda \leq 1$
  - a particle of mass  $1 \text{ kg}$  at the point  $A$
  - a particle of mass  $1 \text{ kg}$  at the point  $C$

[3]

- (c) Show that, as  $\lambda$  varies, the centre of mass of the combined object lies on the curve

$$x = 4y^2 - 2y + k$$

where  $k$  is a constant to be determined.

[3]

### Solution

- (a) Since the three forces form a couple, their resultant force is zero.

Resolving in the  $x$ - and  $y$ -directions:

$$p - q - 3 = 0, \quad p + 2q - 6 = 0$$

Subtracting the first equation from the second gives

$$3q - 3 = 0$$

so

$$q = 1$$

Substituting into  $p - q - 3 = 0$ :

$$p - 1 - 3 = 0 \implies p = 4$$

Now take moments about  $O$ , with anticlockwise positive.

For a force  $F_x \mathbf{i} + F_y \mathbf{j}$  acting at  $(x, y)$ , the moment about  $O$  is

$$xF_y - yF_x$$

So the moments of the three forces are

$$M_1 = 1(2q) - 0(p) = 2q$$

$$M_2 = 0(p) - 1(-q) = q$$

$$M_3 = t(-6) - 1(-3) = 3 - 6t$$

Since the forces form an anticlockwise couple of magnitude  $3 \text{ N m}$ ,

$$M_1 + M_2 + M_3 = 3$$

Hence

$$2q + q + 3 - 6t = 3$$

Using  $q = 1$ :

$$2 + 1 + 3 - 6t = 3$$

Therefore

$$6 - 6t = 3 \implies 6t = 3 \implies t = \frac{1}{2}$$

So

$$p = 4, \quad q = 1, \quad t = \frac{1}{2}$$

- (b) The lamina is light, so its mass is taken to be zero.

Let the centre of mass of the composite body be  $(x, y)$ .

The total mass is

$$2 + 1 + 1 = 4$$

Using  $Mx = m_1x_1 + m_2x_2 + m_3x_3$ ,

$$4x = 2(2\lambda^2) + 1(2) + 1(0) = 4\lambda^2 + 2$$

So

$$x = \lambda^2 + \frac{1}{2}$$

Using  $My = m_1y_1 + m_2y_2 + m_3y_3$ ,

$$4y = 2(\lambda) + 1(0) + 1(1) = 2\lambda + 1$$

So

$$y = \frac{2\lambda + 1}{4}$$

Hence the centre of mass is

$$\left( \lambda^2 + \frac{1}{2}, \frac{2\lambda + 1}{4} \right)$$

(c) From part (b), the centre of mass has coordinates

$$x = \lambda^2 + \frac{1}{2}, \quad y = \frac{2\lambda + 1}{4}$$

Rearranging the expression for  $y$ ,

$$4y = 2\lambda + 1$$

so

$$\lambda = \frac{4y - 1}{2}$$

Substitute this into the expression for  $x$ :

$$\begin{aligned} x &= \left( \frac{4y - 1}{2} \right)^2 + \frac{1}{2} \\ &= \frac{16y^2 - 8y + 1}{4} + \frac{1}{2} \\ &= 4y^2 - 2y + \frac{1}{4} + \frac{1}{2} \\ &= 4y^2 - 2y + \frac{3}{4} \end{aligned}$$

Therefore the locus is

$$x = 4y^2 - 2y + k$$

with

$$k = \frac{3}{4}$$

3. Three particles of masses  $2m$ ,  $5m$  and  $3m$  are placed at the points  $(1, -4)$ ,  $(-3, 2)$  and  $(q, 2q - 1)$  respectively.

The value of  $q$  is such that the distance of the centre of mass of the three particles from the point  $(1, 0)$  is as small as possible.

Find the value of  $q$ .

[7]

### Solution

Let the centre of mass be  $G(\bar{x}, \bar{y})$ .

The total mass is

$$2m + 5m + 3m = 10m$$

Using the centre of mass formula for the coordinates,

$$10m\bar{x} = 2m(1) + 5m(-3) + 3m(q)$$

so

$$\begin{aligned} 10\bar{x} &= 2 - 15 + 3q \\ \bar{x} &= \frac{3q - 13}{10} \end{aligned}$$

Also,

$$10m\bar{y} = 2m(-4) + 5m(2) + 3m(2q - 1)$$

so

$$\begin{aligned} 10\bar{y} &= -8 + 10 + 6q - 3 \\ &= 6q - 1 \\ \bar{y} &= \frac{6q - 1}{10} \end{aligned}$$

Hence

$$G\left(\frac{3q - 13}{10}, \frac{6q - 1}{10}\right)$$

Let the distance from  $G$  to  $(1, 0)$  be  $d$ . Then

$$\begin{aligned} d^2 &= \left(\frac{3q - 13}{10} - 1\right)^2 + \left(\frac{6q - 1}{10}\right)^2 \\ &= \left(\frac{3q - 23}{10}\right)^2 + \left(\frac{6q - 1}{10}\right)^2 \\ &= \frac{(3q - 23)^2 + (6q - 1)^2}{100} \end{aligned}$$

Now expand:

$$\begin{aligned} (3q - 23)^2 + (6q - 1)^2 &= 9q^2 - 138q + 529 + 36q^2 - 12q + 1 \\ &= 45q^2 - 150q + 530 \end{aligned}$$

So

$$d^2 = \frac{45q^2 - 150q + 530}{100}$$

To make  $d$  as small as possible, it is enough to make  $d^2$  as small as possible. Complete the square:

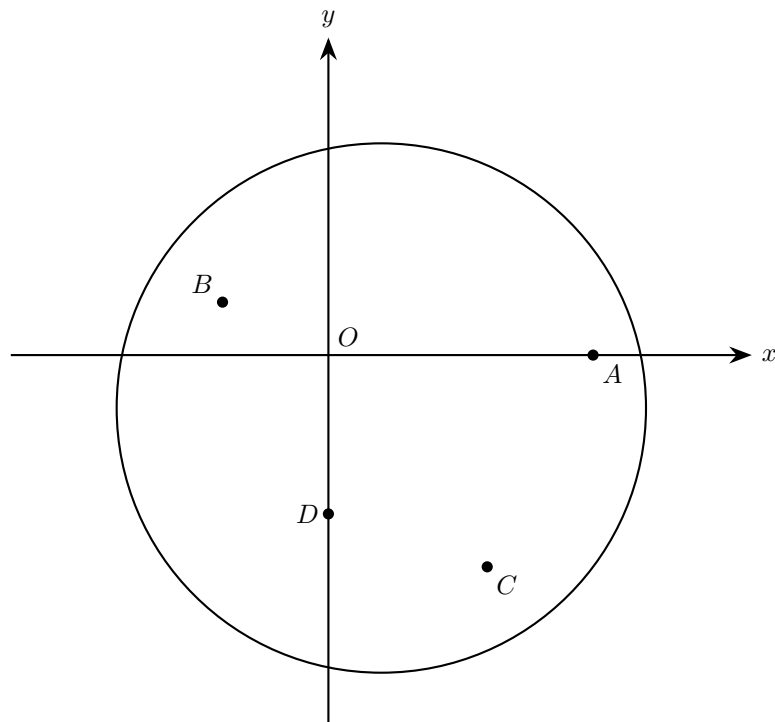
$$\begin{aligned}45q^2 - 150q + 530 &= 45 \left( q^2 - \frac{10}{3}q \right) + 530 \\&= 45 \left[ \left( q - \frac{5}{3} \right)^2 - \frac{25}{9} \right] + 530 \\&= 45 \left( q - \frac{5}{3} \right)^2 - 125 + 530 \\&= 45 \left( q - \frac{5}{3} \right)^2 + 405\end{aligned}$$

Since  $45 \left( q - \frac{5}{3} \right)^2 \geq 0$ , this expression is smallest when

$$q - \frac{5}{3} = 0$$

Therefore,

$$q = \frac{5}{3}$$



4. Four sensors are attached to a circular plate.

The plate is to be modelled as a uniform lamina and the four sensors as four particles.

The diagram shows the lamina, the four particles  $A$ ,  $B$ ,  $C$  and  $D$ , and the coordinate axes.

The lamina has mass 6 kg and its centre of mass is at the point  $(1, -1)$ .

Particle  $A$  has mass 4 kg and is at the point  $(5, 0)$ .

Particle  $B$  has mass 3 kg and is at the point  $(-2, 1)$ .

Particle  $C$  has mass 5 kg and is at the point  $(3, -4)$ .

Particle  $D$  has mass 2 kg and is at the point  $(0, -3)$ .

Find the coordinates of the centre of mass of the system of plate and sensors.

[5]

### Solution

Treat the uniform lamina as a particle of mass 6 kg acting at its centre of mass  $(1, -1)$ .

Let the centre of mass of the whole system be  $(\bar{x}, \bar{y})$ .

The total mass is

$$M = 6 + 4 + 3 + 5 + 2 = 20 \text{ kg}$$

For the  $x$ -coordinate, use the weighted average

$$M\bar{x} = m_1x_1 + m_2x_2 + m_3x_3 + m_4x_4 + m_5x_5$$

so

$$\begin{aligned} 20\bar{x} &= 6(1) + 4(5) + 3(-2) + 5(3) + 2(0) \\ &= 6 + 20 - 6 + 15 + 0 \\ &= 35 \end{aligned}$$

Hence

$$\bar{x} = \frac{35}{20} = \frac{7}{4}$$

For the  $y$ -coordinate,

$$M\bar{y} = m_1y_1 + m_2y_2 + m_3y_3 + m_4y_4 + m_5y_5$$

so

$$\begin{aligned} 20\bar{y} &= 6(-1) + 4(0) + 3(1) + 5(-4) + 2(-3) \\ &= -6 + 0 + 3 - 20 - 6 \\ &= -29 \end{aligned}$$

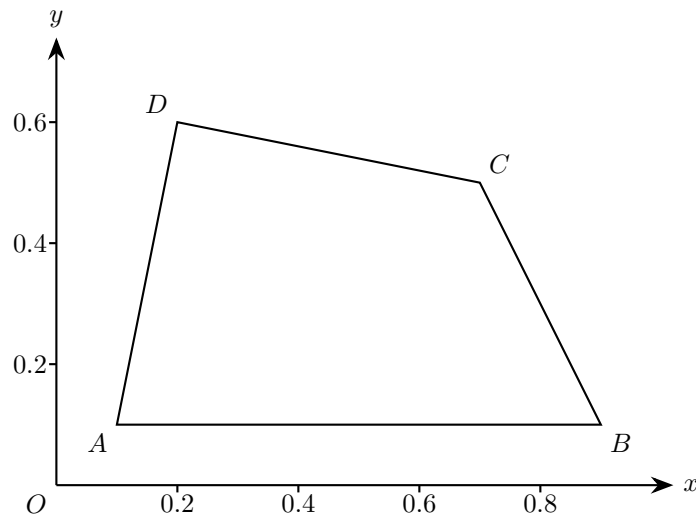
Hence

$$\bar{y} = \frac{-29}{20}$$

Therefore the centre of mass of the system is

$$\left(\frac{7}{4}, -\frac{29}{20}\right)$$

In decimal form, this is  $(1.75, -1.45)$ .



5. Particles  $A$ ,  $B$ ,  $C$  and  $D$  of masses 1 kg, 2 kg, 3 kg and 4 kg respectively are joined by light rigid rods to form the quadrilateral frame shown in the diagram. The particles are at the points  $A(0.1, 0.1)$ ,  $B(0.9, 0.1)$ ,  $C(0.7, 0.5)$  and  $D(0.2, 0.6)$ , where coordinates are measured in metres.

$G$ , the centre of mass of the frame, has coordinates  $(\bar{x}, \bar{y})$ .

- (a) Show that  $\bar{y} = 0.42$ . Find  $\bar{x}$ . [3]
- (b) Using your answer to part (a), explain why the frame cannot be in equilibrium in a horizontal plane if it is supported by a single peg touching the frame at one point. [1]

### Solution

- (a) The total mass of the frame is

$$M = 1 + 2 + 3 + 4 = 10$$

Using the centre of mass formula for particles,

$$M\bar{x} = \sum mx, \quad M\bar{y} = \sum my$$

For the  $y$ -coordinate,

$$10\bar{y} = 1(0.1) + 2(0.1) + 3(0.5) + 4(0.6)$$

$$10\bar{y} = 0.1 + 0.2 + 1.5 + 2.4 = 4.2$$

$$\bar{y} = \frac{4.2}{10} = 0.42$$

So  $\bar{y} = 0.42$ , as required.

For the  $x$ -coordinate,

$$10\bar{x} = 1(0.1) + 2(0.9) + 3(0.7) + 4(0.2)$$

$$10\bar{x} = 0.1 + 1.8 + 2.1 + 0.8 = 4.8$$

$$\bar{x} = \frac{4.8}{10} = 0.48$$

Hence the centre of mass is

$$G = (\bar{x}, \bar{y}) = (0.48, 0.42)$$

- (b) For equilibrium on a single peg, the peg must be vertically below the centre of mass. So in the horizontal plane, the peg would have to touch the frame at  $G = (0.48, 0.42)$ . But this point is not on the frame itself; it lies inside the quadrilateral, not on any of its sides. Therefore the frame cannot be in equilibrium if it is supported by a single peg touching the frame at one point.

6. Four particles of mass  $m$ ,  $2m$ ,  $4m$  and  $\lambda m$  are placed at points with coordinates  $(0, 0)$ ,  $(2, 3)$ ,  $(1, -1)$  and  $(4, 2)$  respectively. The centre of mass of the system is at  $(2, k)$ .

(a) Show that  $\lambda = 3$ . [4]

(b) Calculate the value of  $k$ . [3]

### Solution

(a) Let the total mass be  $M$ . Then

$$M = m + 2m + 4m + \lambda m = (7 + \lambda)m$$

Using the  $x$ -coordinate of the centre of mass, with centre  $(2, k)$ ,

$$M \cdot 2 = m \cdot 0 + 2m \cdot 2 + 4m \cdot 1 + \lambda m \cdot 4$$

So

$$2(7 + \lambda)m = 0 + 4m + 4m + 4\lambda m$$

$$2(7 + \lambda) = 8 + 4\lambda$$

$$14 + 2\lambda = 8 + 4\lambda$$

$$6 = 2\lambda$$

$$\lambda = 3$$

Hence  $\lambda = 3$ .

(b) Now use the  $y$ -coordinate of the centre of mass. Since  $\lambda = 3$ ,

$$M = (7 + 3)m = 10m$$

Also,

$$M \cdot k = m \cdot 0 + 2m \cdot 3 + 4m \cdot (-1) + 3m \cdot 2$$

So

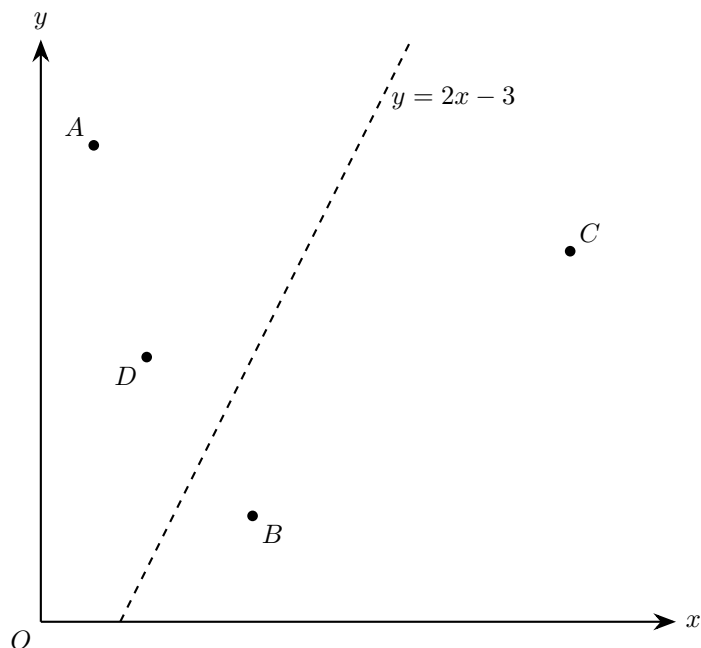
$$10mk = 0 + 6m - 4m + 6m$$

$$10mk = 8m$$

$$k = \frac{8}{10}$$

$$k = \frac{4}{5}$$

So the value of  $k$  is  $\frac{4}{5}$ .



7. The diagram shows four particles,  $A$ ,  $B$ ,  $C$  and  $D$ , fixed in a horizontal plane which contains the  $x$ - and  $y$ -axes.

Particle  $A$  has mass 3 kg and is attached at the point  $(1, 9)$ .  
 Particle  $B$  has mass 4 kg and is attached at the point  $(4, 2)$ .  
 Particle  $C$  has mass 5 kg and is attached at the point  $(10, 7)$ .  
 Particle  $D$  has mass  $m$  kg and is attached at the point  $(2, 5)$ .

The centre of mass of the four particles lies on the line  $y = 2x - 3$ .  
 Find the value of  $m$  and the coordinates of the centre of mass.

[6]

### Solution

Let the centre of mass be  $(\bar{x}, \bar{y})$ .

The total mass is

$$M = 3 + 4 + 5 + m = 12 + m$$

Using the centre of mass formula for coordinates,

$$M\bar{x} = 3(1) + 4(4) + 5(10) + m(2)$$

so

$$\begin{aligned} (12 + m)\bar{x} &= 3 + 16 + 50 + 2m \\ &= 69 + 2m \end{aligned}$$

Hence

$$\bar{x} = \frac{69 + 2m}{12 + m}$$

Similarly,

$$M\bar{y} = 3(9) + 4(2) + 5(7) + m(5)$$

so

$$\begin{aligned} (12 + m)\bar{y} &= 27 + 8 + 35 + 5m \\ &= 70 + 5m \end{aligned}$$

Hence

$$\bar{y} = \frac{70 + 5m}{12 + m}$$

Since the centre of mass lies on the line  $y = 2x - 3$ ,

$$\frac{70 + 5m}{12 + m} = 2 \left( \frac{69 + 2m}{12 + m} \right) - 3$$

Multiply through by  $12 + m$ :

$$\begin{aligned} 70 + 5m &= 2(69 + 2m) - 3(12 + m) \\ &= 138 + 4m - 36 - 3m \\ &= 102 + m \end{aligned}$$

Therefore

$$\begin{aligned} 70 + 5m &= 102 + m \\ 4m &= 32 \\ m &= 8 \end{aligned}$$

Now substitute  $m = 8$ . Then the total mass is

$$12 + 8 = 20$$

So

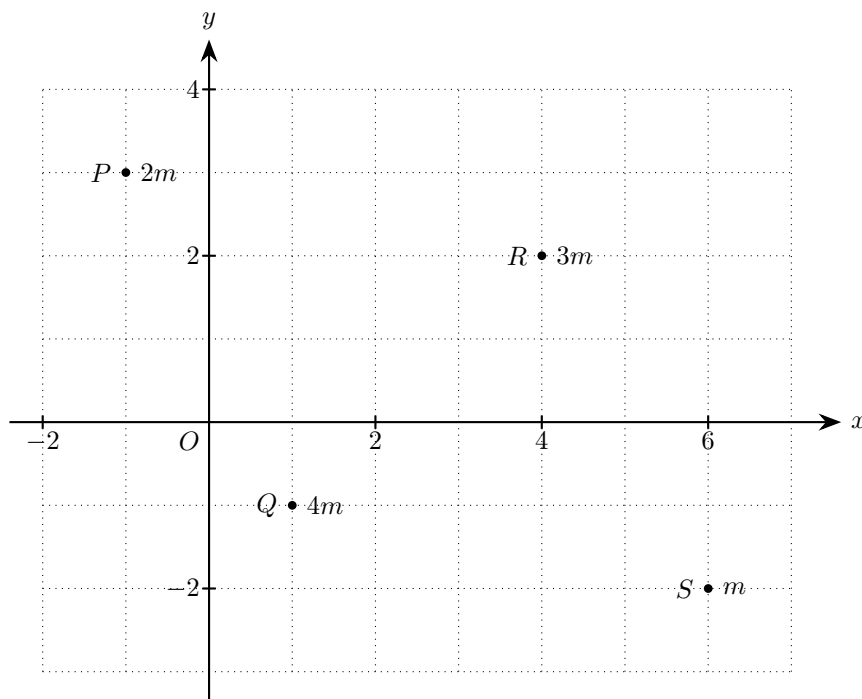
$$\begin{aligned} \bar{x} &= \frac{69 + 2(8)}{20} = \frac{85}{20} = \frac{17}{4} \\ \bar{y} &= \frac{70 + 5(8)}{20} = \frac{110}{20} = \frac{11}{2} \end{aligned}$$

Therefore,

$$m = 8$$

and the centre of mass is

$$\left( \frac{17}{4}, \frac{11}{2} \right)$$



8. The diagram shows a system of four particles of masses  $2m$ ,  $4m$ ,  $3m$  and  $m$  situated in the  $x$ - $y$  plane at the points  $P(-1, 3)$ ,  $Q(1, -1)$ ,  $R(4, 2)$  and  $S(6, -2)$  respectively.

Find the coordinates of the centre of mass of the system.

[5]

### Solution

Let the centre of mass be  $(\bar{x}, \bar{y})$ .

First find the total mass:

$$M = 2m + 4m + 3m + m = 10m$$

For the  $x$ -coordinate of the centre of mass, use the weighted average:

$$M\bar{x} = 2m(-1) + 4m(1) + 3m(4) + m(6)$$

So

$$10m\bar{x} = -2m + 4m + 12m + 6m = 20m$$

Hence

$$\bar{x} = 2$$

For the  $y$ -coordinate of the centre of mass:

$$M\bar{y} = 2m(3) + 4m(-1) + 3m(2) + m(-2)$$

So

$$10m\bar{y} = 6m - 4m + 6m - 2m = 6m$$

Hence

$$\bar{y} = \frac{3}{5}$$

Therefore the coordinates of the centre of mass are  $(2, \frac{3}{5})$ .

9. Four particles of masses  $m$ ,  $2m$ ,  $3m$  and  $km$  are positioned at the points with coordinates  $(-a, a)$ ,  $(2a, -2a)$ ,  $(0, 0)$  and  $(\mu a, 2\mu a)$  respectively, where  $k$  and  $\mu$  are constants.

The centre of mass of the four particles is at the point with coordinates  $(2a, 3a)$ .

(a) Find the value of  $k$ . [3]

(b) Find the value of  $\mu$ . [3]

### Solution

(a) Let the total mass be

$$M = m + 2m + 3m + km = (6 + k)m$$

Using the formula for the  $x$ -coordinate of the centre of mass,

$$M(2a) = m(-a) + 2m(2a) + 3m(0) + km(\mu a)$$

so

$$(6 + k)m(2a) = -am + 4am + 0 + k\mu am$$

$$2(6 + k) = 3 + k\mu$$

$$12 + 2k = 3 + k\mu$$

$$k\mu = 9 + 2k$$

Using the formula for the  $y$ -coordinate of the centre of mass,

$$M(3a) = m(a) + 2m(-2a) + 3m(0) + km(2\mu a)$$

so

$$(6 + k)m(3a) = am - 4am + 0 + 2k\mu am$$

$$3(6 + k) = -3 + 2k\mu$$

$$18 + 3k = -3 + 2k\mu$$

$$2k\mu = 21 + 3k$$

Now double the equation  $k\mu = 9 + 2k$ :

$$2k\mu = 18 + 4k$$

Equating the two expressions for  $2k\mu$ ,

$$18 + 4k = 21 + 3k$$

$$k = 3$$

(b) Substitute  $k = 3$  into

$$k\mu = 9 + 2k$$

to get

$$3\mu = 9 + 2(3)$$

$$3\mu = 15$$

$$\mu = 5$$